Cross-View Object Geo-Localization in a Local Region With Satellite Imagery

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***Abstract*— Cross-view geo-localization is a critical task in various applications, such as smart city management and disaster monitoring. Current methods typically divide a satellite image into patches and use these patches to identify the geographic location of a query image. However, these methods can only provide the location of an image rather than the location of a specific object of interest. This makes it difficult to link these methods to GeoDatabases to obtain detailed information about a target object, such as its name and construction time. To overcome this limitation, we propose a novel problem of cross-view object geo-localization (CVOGL) in a local region with high-resolution satellite images. This problem includes two main challenges: accurately identifying the location of an object and distinguishing the target object from others in satellite images. To address these challenges, we present a new Detection-based Geo-localization method called DetGeo, which consists of an object detection-based framework with a two-branch encoder and a query-aware cross-view fusion module. DetGeo uses cross-view images as input to the detector to provide object-level geo-localization. The fusion module employs cross-view spatial attention to focus on relevant areas of target objects during cross-view feature fusion. To evaluate our method, we constructed a new CVOGL dataset, which comprises ground- or drone-view images as query images and satellite-view images as geo-tagged reference images. Comprehensive experiments are conducted to demonstrate the effectiveness of our method on CVOGL (https://github.com/sunyuxi/DetGeo).**

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I. Introduction

CROSS-VIEW image geo-localization [1], [2], [3], [4], [5], [6] refers to the technique of determining the geographic location of a query image by finding the most similar image from a database of geo-tagged reference images. The query images can be captured from various devices, such as phone cameras and drones, while the reference images are typically high-resolution satellite images that are annotated with geographical tags [7], [8]. By utilizing the geo-tags of the retrieved images, it becomes possible to make accurate predictions about the location of the target area. This technique is widely used in various fields, such as drone navigation [4], [9], smart city management [10], [11], [12], and disaster monitoring [13], [14]. However, due to drastic viewpoint changes between the query images and database images, cross-view geo-localization requires learning a discriminative image representation to effectively match these images.

Cross-view image geo-localization methods can generally be classified into two categories: metric learning-based methods [1], [2], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25] and polar transform-based methods [26], [27], [28], [29], [30], [31], [32], [33]. The former methods aim to directly learn viewpoint-invariant features, neglecting geometric configurations, and utilize image retrieval techniques to match cross-view images. For example, Hu et al. [16] employ a Siamese neural network and a weighted soft margin ranking loss to learn a viewpoint-invariant representation. On the other hand, geometry-based methods focus on reducing the viewpoint discrepancy between ground- and satellite-view images by employing geometric configurations, such as orientation information. For example, Shi et al. [27] employ a polar transform to warp a satellite image such that ground- and satellite-view domains are aligned geometrically. These cross-view geo-localization methods have demonstrated good performance in general. However, current cross-view geo-localization methods are primarily designed to identify the position of an image, rather than the location of an object. As a result, they cannot be directly linked to an additional GeoDatabase (e.g., OpenBuildingMap [34]) to obtain detailed information about an object, which may limit their practicality in some real-world scenarios.

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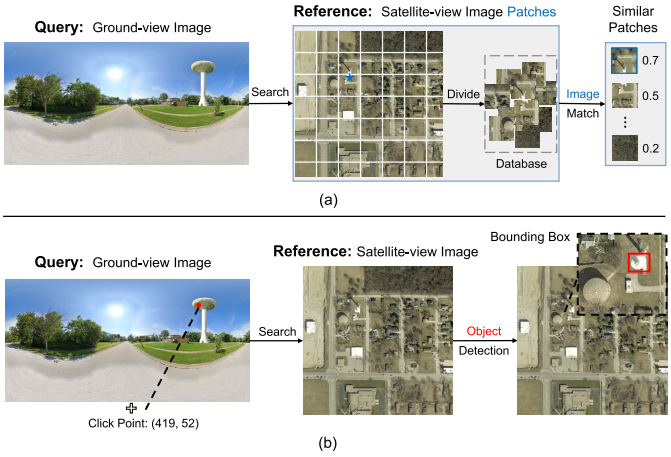


Fig. 1. Comparison of existing cross-view image geo-localization and our CVOGL. First, existing methods only find the approximate location of a query image, which may contain multiple objects at varying positions. Our method can precisely identify the location of a specific object selected by a click point in the query image. Second, existing methods for geo-localization typically rely on matching satellite image patches, which may contain multiple objects or only cover parts of an object. Our method employs the entire satellite image to accurately locate a single object. (a) Cross-view image geo-localization. (b) CVOGL (ours).

Cross-view geo-localization of objects plays a vital role in various applications, such as smart city management [10], [11], [12] and disaster monitoring [13], [14]. For example, in the context of city management, a query image may show a water tower of interest or with issues, as depicted in Fig. 1. In such scenarios, city workers not only need to know the tower's location but also other details, such as its name and construction time, typically stored in a GeoDatabase [35]. However, current cross-view image geo-localization methods can only provide a nearby location for the object, which is not enough to directly access its details from the GeoDatabase. As a result, city workers must manually locate the water tower on a map and query the GeoDatabase for its detailed information. The process is time-consuming and can even be impossible if the object is not documented on the maps [36]. Therefore, accurate cross-view geo-localization of objects is crucial in these situations.

However, current cross-view image geo-localization methods have limitations in locating specific objects for two main reasons. First, as illustrated in Fig. 1(a), a query image may contain multiple objects at different locations, but current image geo-localization methods can only determine the approximate location of the query image. The location where the query image was captured does not necessarily correspond to the location of these objects, resulting in current methods being ineffective for object localization. Second, as depicted in Fig. 1(a), since most satellite image patches cannot capture a single object, the geo-tags of these image patches usually do not correspond to the object's location. Current methods typically divide an original satellite image into several patches before cross-view geo-localization. However, objects in satellite images vary greatly in size, even within the same class [37], [38], and each patch may contain multiple objects or only cover part of an object. As a result, it is impossible to design a perfect division solution, which would ensure that each image patch's location corresponds to the object's location.

To address these limitations, we propose a novel task of cross-view object geo-localization (CVOGL), as shown in Fig. 1(b), which directly uses a high-resolution satellite image to detect the location of a specific object indicated by a click point in a query image. This task focuses on a local region because certain users, such as city workers, often require information about a specific area rather than the entire global region. As far as we know, there are no datasets currently available that are suitable for exploring the particular problem that we are addressing. Therefore, we constructed a CVOGL dataset, which consists of 12 478 instances, including 5836 high-resolution satellite images, 5279 ground-view images, and 5279 drone aerial-view images. Each ground-view and drone aerial-view image in the dataset contains a query object marked by a click point. In addition, each satellite image is annotated with a bounding box (bbox) indicating the geographic location of the corresponding query object. The CVOGL dataset can be used to investigate two CVOGL tasks: ground-to-satellite object localization and drone-to-satellite object localization. Both tasks aim to locate objects based on cross-view images and are essentially the same.

The CVOGL task presents two main challenges. First, accurately identifying the location of objects is difficult using existing geo-localization frameworks. This is because these frameworks mainly rely on image retrieval methods [39], [40], [41], [42], [43], [44] that provide only image-level localization, which is insufficient for this task. Therefore, it is crucial to design a framework that can provide object-level geo-localization. Second, satellite images are used as reference images for this task, but it is challenging to distinguish a target object from other objects in these images. This is because satellite images cover a large imaging range, containing numerous objects. Moreover, query and reference images are multimodal data. Usually, there are significant visual differences and a large domain gap between multimodal data [45], [46]. Therefore, comprehending these cross-view images and precisely locating objects within them is challenging.

To address these challenges, we propose a novel Detection-based Geo-localization method called `DetGeo`. First, our method utilizes an object DetGeo framework with a two-branch encoder to locate an object depicted by a query image from a satellite-view reference image. The query image can be either a ground-view image or a drone-view image. The two-branch encoder independently extracts visual features for each view image, enabling us to predict the bbox of a target object by inputting cross-view images into an object detector. Our method provides object-level cross-view geo-localization. Second, we design a query-aware cross-view fusion module that uses cross-view spatial attention to effectively fuse images. This mechanism allows the fusion module to focus on the most relevant areas of the reference images that are most likely to contain the target object. By dynamically adjusting the attention weights based on the query image, the module can learn and fuse query and reference images for more precise object geo-localization. We evaluated our proposed method

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on the CVOGL dataset and demonstrated its effectiveness for CVOGL.

The contributions of this article can be summarized as follows.

1) We introduce a novel CVOGL problem that focuses on localizing objects within a local region using cross-view images. To the best of our knowledge, this problem has not been explored previously, presenting a new challenge to the remote sensing community.
2) We constructed a new dataset called CVOGL, comprising ground- or drone-view images as query images and satellite images as reference images. Our dataset is the first of its kind to be used in studying the CVOGL problem.
3) We propose a new object `DetGeo` method, which predicts the exact object location using cross-view images. Our method includes a query-aware cross-view fusion module that employs cross-view spatial attention to effectively learn and fuse cross-view images. Comprehensive experiments demonstrate the effectiveness of our method on the CVOGL dataset.

The rest of this article is organized as follows. Section II provides a brief review of the related work in cross-view geo-localization. Section III describes the problem definition and dataset construction. Section IV presents a detailed description of our method. In Section V, we present the results of our comprehensive experiments on CVOGL. Finally, we conclude this article with some remarks in Section VI.

II. Related Work

A. Cross-View Geo-Localization Datasets

Several datasets [7], [15], [26], [47], [48], [49], [50], [51], [52], [53], [54], [55] have been proposed to evaluate cross-view geo-localization tasks. For example, CVUSA [15], [47] and CVACT [26] use street-view images as query images to find geographic locations from GPS-tagged very-high-resolution optical satellite images. GRAL [52] employs aerial LIDAR depth images as reference images to estimate the location of a ground-view image. SUES-200 [55] focuses on identifying the localization of drone-view images with different heights by matching satellite images. These datasets mainly contain two view images. When images involve more views or spectrums, tasks may become more challenging [56], [57]. Zheng et al. [7] introduce a multiview University-1652 dataset that includes drone-, satellite-, and ground-view images to boost viewpoint-invariant feature learning. In these datasets (e.g., CVUSA and GRAL), query and reference images are taken at the same location but at different times, which increases the difficulty of image geo-localization. Thus, Samano et al. [51] use a 2-D cartographic map as a reference source to reduce the invariance challenge. In addition, the aforementioned datasets mainly consider one-to-one retrieval, where a query ground-level image corresponds to a reference aerial or drone image. Zhu et al. [54] introduce a VIGOR dataset beyond one-to-one retrieval, where multiple reference images partially cover the same query image.

To the best of our knowledge, current cross-view geo-localization datasets only consider the location of a query image, rather than the location of a specific object within the query image. To address this limitation, we have constructed a novel CVOGL dataset that includes ground- or drone-view images as query images and very-high-resolution satellite images as reference images.

B. Cross-View Geo-Localization Methods

Existing methods for cross-view geo-localization can be roughly categorized into metric learning- and geometry-based methods. Metric learning-based methods [1], [2], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25] directly learn viewpoint-invariant features for matching ground- and aerial-view images without considering geometric configurations. For example, Workman et al. [15] first introduce CNNs to learn a joint semantic feature representation for cross-view image geo-localization. Hu et al. [16] adopt a Siamese neural network and a weighted soft margin ranking loss to learn a viewpoint-invariant representation. With the Siamese architecture, Cai et al. [17] design a reweighting triplet loss to focus on informative hard exemplars during model training. Shi et al. [18] develop a cross-view feature transport technique to align the visual features between ground- and aerial-view images. Yang et al. [19] apply Transformer to learn the global dependencies of cross-view images to reduce the visual ambiguities in cross-view matching. Lu et al. [1] propose a step-adaptive iterative refinement network for cross-view image geo-localization, which adaptively refines the learning process in multiple steps and dynamically adjusts the number of refinements using an adaptive step estimation mechanism. Zhu et al. [2] introduce an end-to-end network architecture composed of ResNetX and GeoCaps modules, where the GeoCaps module utilizes capsule networks and dynamic routing-by-agreement to model the spatial hierarchy information of scene objects. Zhu et al. [20] use a Transformer to learn the geometric correspondence between two views and design an attention-guided nonuniform cropping method to focus on important regions when learning cross-view representations. Sun et al. [58] propose an end-to-end geo-localization framework to address the domain gap between drone- and satellite-view images. Their method establishes semantic consistency between multisource and multiview image pairs by capturing both local and global information.

Geometry-based methods [26], [27], [28], [29], [30], [31], [32], [33] explicitly reduce the viewpoint discrepancy between ground- and aerial-view images by using geometric configurations such as orientation information. For example, Liu and Li [26] explicitly encode the orientation of each pixel of the image to generate more discriminative features. Shi et al. [27] apply a polar transform to warp an aerial-view image such that ground- and aerial-view image domains are aligned geometrically. Similarly, Shi et al. [28] apply a polar transform to aerial-view images and design a dynamic similarity matching module to account for orientation so that the cross-view domain gap is reduced when measuring the feature similarity of image pairs. Regmi and Shah [29] synthesize an

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aerial-view image from a ground-view image using conditional GANs and fuse the synthesized image with the original ground image to reduce visual ambiguities in cross-view geo-localization. Lu et al. [30] learn the cross-view representation by generating panoramic street-view images consistent with a given satellite image, which explicitly incorporates the geometric configuration of ground objects from the satellite view. Toker et al. [31] further apply a GAN network on top of polar-transformed aerial-view images and design a multitask architecture to jointly train image synthesis and cross-view matching. Guo et al. [33] propose a geometric and scene information fusion model for cross-view geo-localization, which learns the geometric information of cross-view image scenes via log-polar transformation and spatial attention aggregation. Zhang et al. [59] introduce GeoDTR to enhance the performance of cross-view geo-localization on cross-area benchmarks. GeoDTR extracts the spatial configuration of visual feature layouts by explicitly disentangling geometric information from raw features. It also learns the spatial correlations among visual features from aerial and ground pairs using a novel geometric layout extractor module.

As far as we know, existing cross-view geo-localization methods mainly focus on image-level localization rather than object-level localization. In this article, we propose a novel method to find the exact location of an object from cross-view images.

III. Problem and Dataset Construction

First, we define the problem statement of our proposed task in Section III-A. Next, we provide a detailed description of our data collection pipeline, including cross-view image collection in Section III-B and object annotation in Section III-C. Finally, in Section III-D, we conducted a statistical analysis of the collected dataset.

A. Problem Statement

Given a local region of interest, our objective is to find the location of an object in this region, where the object is shown in a query image and the region is represented by a geo-tagged reference image. To achieve this objective, a dataset of n instances $\mathcal{X} = \{x_i\}_{i=1}^{n} = \{(q_i, r_i, p_i, b_i)\}_{i=1}^{n}$ needs to be constructed, where q_i and r_i denote a query image and a reference image, respectively. p_i is a click point and is used to specify a query object in q_i. The corresponding geographic location of the query object in r_i is b_i. In this article, we use a bbox to represent the location, i.e., $b_i = (x_{b_i}, y_{b_i}, w_{b_i}, h_{b_i})$, where (x_{b_i}, y_{b_i}), w_{b_i}, and h_{b_i} are the center coordinate, width, and height of the bbox, respectively. Finally, the problem is formulated as $(q_i, r_i, p_i) \mapsto b_i$.

B. Cross-View Image Collection

To construct the dataset as depicted in Fig. 2, we collected ground-view and drone aerial-view images as query images and satellite-view images as reference images. Google Street-View Static API[1] was used to download the ground-view images, which are sized at 256 × 512 pixels. Drone aerial-view images, measuring 256 × 256 pixels, were obtained from vexcel imaging.[2] Satellite-view images, which represent various local regions in the United States, were collected from multiple sensors and platforms, such as Microsoft Bing Maps and ArcGIS Maps [60]. The images are all sized at 1024 × 1024 pixels, but they vary in resolution. The coverage of each image ranges from 0.09 to 1 km^2, providing a broad representation of local regions across the United States.

TABLE I
Statistics of the CVOGL Dataset

Task	Split	Instances	Query	Reference
Ground → Satellite	total	6,239	5,279	5,836
	training	4,343	3,674	4,063
	validation	923	793	863
	test	973	812	910
Drone → Satellite	total	6,239	5,279	5,836
	training	4,343	3,674	4,063
	validation	923	793	863
	test	973	812	910

C. Object Annotation

To construct the dataset, we manually annotated the locations of certain objects in the cross-view images. We mainly focus on common objects in satellite images, such as buildings, bridges, roundabouts, baseball fields, and storage tanks. These objects are typically used in remote sensing object detection datasets, such as DOTA [37]. We obtain the boundaries of these objects using OpenStreetMap.[3] An example of a building annotated in OpenStreetMap is shown in Fig. 3. To annotate these objects, we followed these steps.

First, we recorded the location information, including the latitude and longitude of the bottom left and top right corners, for each downloaded satellite image. Second, we randomly selected some objects from these satellite images and obtained their bboxs from OpenStreetMap. We manually checked these bboxs and discarded any incorrect ones. Third, we selected street-view or drone aerial-view images near these bboxs. Fourth, we selected the query object that corresponds to the annotated object in the satellite image by clicking on the street-view images and drone aerial-view images. If we could not find the same object, we discarded the street-view and drone aerial-view images. Finally, we saved the street-view and drone aerial-view images along with their click points and the corresponding satellite images with bboxs.

D. Data Analysis

After the data processing described above, we constructed the CVOGL dataset, which comprises 12 478 instances with 5836 high-resolution satellite images, 5279 ground-view images, and 5279 drone aerial-view images, as shown in Table I. Several examples from the dataset are depicted

[1] https://developers.google.com/maps/documentation/streetview/request-streetview

[2] https://www.vexcel-imaging.com/vdp/

[3] https://www.openstreetmap.org

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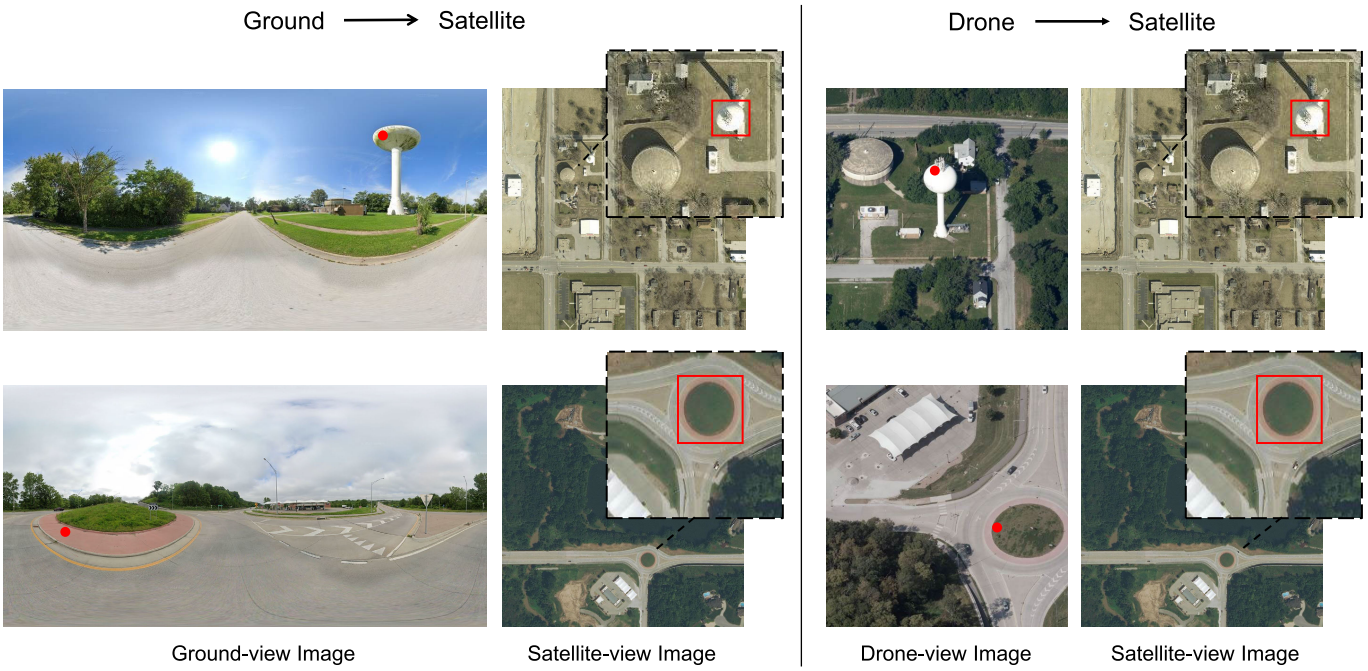


Fig. 2. These examples are taken from CVOGL and demonstrate ground-to-satellite object geo-localization and drone-to-satellite object geo-localization. (Left) Each row shows a ground-view image with a click point, as well as a satellite-view image with a corresponding bbox. (Right) Each row displays a drone aerial-view image with a click point and a satellite-view image with a bbox.

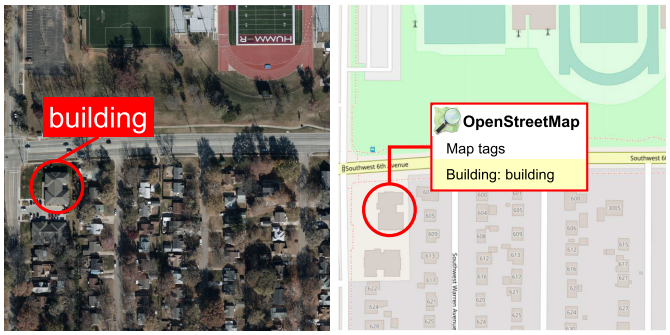


Fig. 3. Annotation example of OpenStreetMap data.

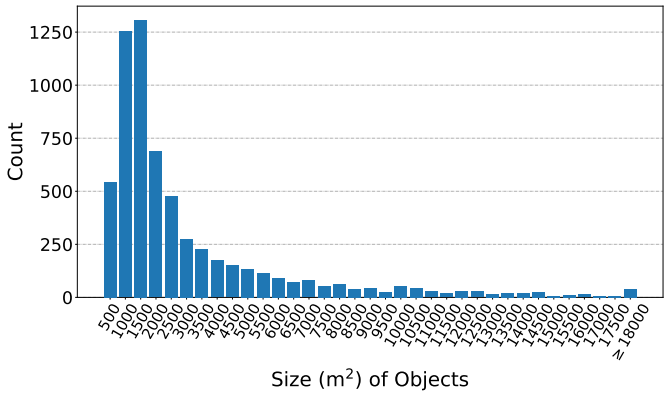


Fig. 4. Size (m^2) distribution of objects in the CVOGL dataset.

in Fig. 2. The CVOGL dataset can be employed to investigate two CVOGL tasks: Task 1 (Ground → Satellite) utilizes ground-view images as query images and high-resolution satellite images as reference images, while Task 2 (Drone → Satellite) employs drone aerial-view images as query images and high-resolution satellite images as reference images. Both tasks aim to locate objects based on cross-view images and are essentially the same. We partitioned the dataset into two subdatasets for the two tasks. Then, in each subdataset, we further split it into training, validation, and test sets, as shown in Table I. Importantly, in each subdataset, a query object may appear in different satellite images because some satellite images contain overlapping regions. Therefore, we split the subdataset by query images to ensure that the same query objects are included in the same data split.

CVOGL contains various objects from different classes, such as buildings and baseball fields. These objects vary significantly in size, as demonstrated in Fig. 4, which illustrates the distribution of object sizes in CVOGL. The dataset encompasses objects with areas ranging from less than 500 to over 18 000 m^2. The wide range of object sizes makes it challenging to locate objects accurately from cross-view images using existing methods. It is impossible for existing methods to design a perfect solution that ensures each image patch contains only one object, and the patch location corresponds exactly to the object's location. However, our proposed method can handle objects of different sizes and accurately locate them in cross-view images.

CVOGL contains various satellite images that capture different local regions with varying coverage ranges. Detecting the location of objects in larger regions is typically more challenging than in smaller ones due to the increased number of objects present. Table II provides the statistic of the number of regions with different sizes, revealing that all data splits include regions with diverse coverage extents, such as 300 × 300 and 1024 × 1024. This poses a significant challenge for CVOGL. Nevertheless, our approach can leverage cross-view spatial attention to identify key areas and

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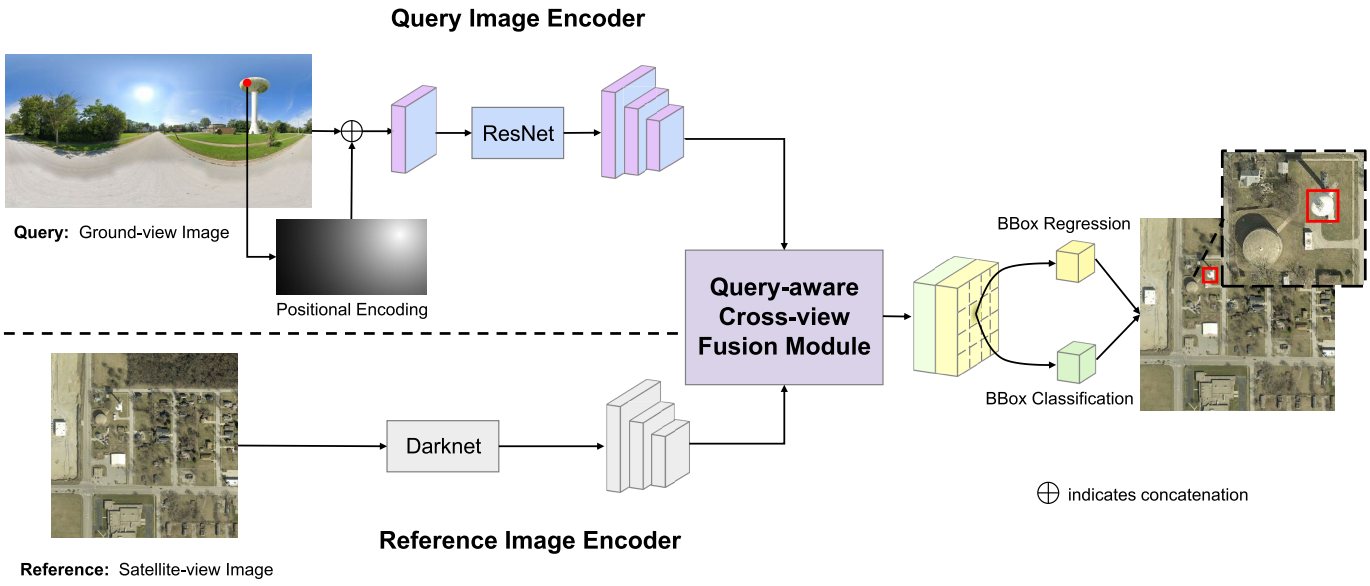


Fig. 5. Overall architecture of our proposed DetGeo consists of three main components: a query image encoder, a reference image encoder, and a query-aware cross-view fusion module. The query image can either be a ground-view or drone aerial-view image. To represent a click point, we use positional encoding, which is a positional matrix of the same size as the corresponding query image; bbox refers to a bounding box.

TABLE II
SIZE (m^2) DISTRIBUTION OF REGIONS IN THE CVOGL DATASET

Task	Split	300×300	512×512	1024×1024
Ground → Satellite	training	2,547	1,127	669
	validation	548	245	130
	test	545	267	161
Drone → Satellite	training	2,547	1,127	669
	validation	548	245	130
	test	545	267	161

effectively address the challenge of satellite images with varying coverage ranges.

IV. METHODOLOGY

A. Overview

Our goal is to precisely locate an object in cross-view images, typically represented by a bbox. To achieve this, we propose a novel detection-based CVOGL method, called DetGeo, as shown in Fig. 5. DetGeo comprises a two-branch encoder network and a query-aware cross-view fusion module. The network extracts visual features using a query image encoder and a reference image encoder. The query image encoder can be applied to either ground-view or drone aerial-view images, while the reference image encoder can be applied to satellite-view images. The query-aware cross-view fusion module combines these extracted features using cross-view spatial attention to generate fused features, which are used to predict the bbox. Finally, we develop an objective function to train DetGeo.

B. Query Image Encoder

In CVOGL, a query object is selected by clicking on a point in a query image. Therefore, the query image encoder must not only extract features from query images but also learn to recognize the click points to identify which objects are being queried. To accomplish this, we first encode the click point p'_k of the kth query image q_k as a positional matrix, where each element is determined by calculating the distance between the clicked point and each point in the query image. The encoding process is given as follows:

$$\mathbf{P}_k(i,j) = (1 - \|(z_k(i,j) - p_k)\|_2/\text{const})^2 \tag{1}$$

where $\mathbf{P}_k$ denotes the positional matrix for p_k. $\mathbf{P}_k(i,j)$ is the entry in the ith row and the jth column of the matrix. $z_k(i,j)$ represents a point with a coordinate (i,j), and $\|\cdot\|_2$ is the Euclidean distance between $z_k(i,j)$ and the click point p_k. The diagonal length of the query image is represented by the constant const. This matrix is visualized as a positional encoding image in Fig. 5, where the brightest pixel indicates the query object.

Next, we concatenate the positional matrix with its corresponding query image. ResNet, known for its exceptional performance in visual representation, is commonly employed as the feature extractor for ground-view images and drone-view images [2], [17], [58]. Therefore, we utilize the ResNet18 [61] architecture with multiple convolutional layers to extract the features of the concatenated data. The process is defined as follows:

$$\mathbf{F}_k^q = \text{ReLU}(\text{BN}(\text{Conv}(\phi(q_k \| \mathbf{P}_k)))) \tag{2}$$

where ϕ denotes the ResNet18 network and Conv is the convolutional network. BN and ReLU represent a batch normalization layer and an active function.

Finally, we obtain the feature map generated by the query image encoder, denoted as $\mathbf{F}_k^q \in \mathbb{R}^{C_q \times H_q \times W_q}$. Here, C_q, H_q, and W_q represent the number of channels, height, and width of the feature map, respectively.

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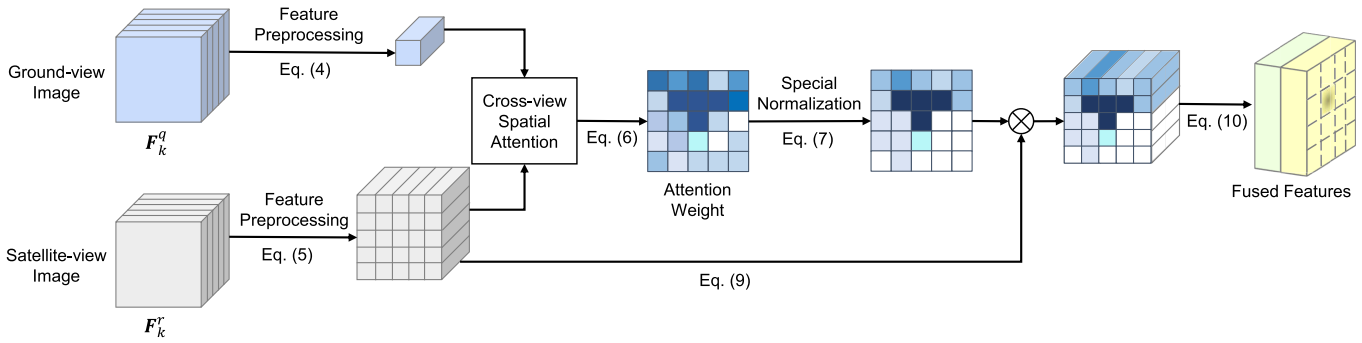


Fig. 6. Structure of the query-aware cross-view fusion module.

C. Reference Image Encoder

CVOGL aims to identify a location from a satellite image. The location is indicated by a bbox utilized in object detection. As mentioned in [37] and [62], Darknet is commonly employed as the backbone network in one-stage object detection methods, such as the YOLO series. Darknet incorporates a feature pyramid network to extract multiscale features from satellite images, enabling effective representation of bboxs for objects of various sizes. Thus, in the reference image encoder, we use an object detection network, such as Darknet [63], and several convolutional networks to extract visual features for a reference image, which is defined as follows:

$$\mathbf{F}_k^r = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(\varphi(r_k)))) \tag{3}$$

where $\mathbf{F}_k^r \in \mathbb{R}^{C_r \times H_r \times W_r}$ is the feature map extracted by the reference image encoder. C_r is the number of channels of $\mathbf{F}_k^r$, which is the same as the number of channels of $\mathbf{F}_k^q$. H_r and W_r denote the height and the width of $\mathbf{F}_k^r$, respectively.

D. Query-Aware Cross-View Fusion Module

In the CVOGL task, the goal is to determine the geographic location of a query object using a query image and a reference image. To achieve this goal, we need to fuse the visual features extracted from the query and reference images. However, the reference image typically contains numerous objects and has significant visual differences from the query image, making it challenging to fuse these visual features to locate the query object accurately. Therefore, we design a query-aware cross-view fusion module to effectively combine the visual features. The structure of the fusion module is depicted in Fig. 6. This module assigns weights to different objects in the reference image based on the query image and pays more attention to the target object. The fusion features generated by the module can more accurately locate the geographic location of the query object.

The fusion module utilizes cross-view spatial attention to combine the visual features of different views (i.e., $\mathbf{F}_k^q$ and $\mathbf{F}_k^r$). As shown in Fig. 6, to generate the fused features, we first preprocess the query image's original feature $\mathbf{F}_k^q \in \mathbb{R}^{C_q \times H_q \times W_q}$ to produce the normalized feature $\hat{\mathbf{F}}_k^q \in \mathbb{R}^{C_q}$, where the mth element of $\hat{\mathbf{F}}_k^q$ is defined as follows:

$$\hat{\mathbf{f}}_k^q(m) = \frac{\sum_{i=1}^{H_q}\sum_{j=1}^{W_q} f_k^q(m,i,j)/\left(H_q W_q\right)}{\sqrt{\sum_{l=1}^{C_q}\left(\sum_{i=1}^{H_q}\sum_{j=1}^{W_q} f_k^q(l,i,j)/\left(H_q W_q\right)\right)^2}} \tag{4}$$

where $f_k^q(m,i,j)$ indicates the (m,i,j)th element of $\mathbf{F}_k^q$. Equation (4) calculates the pooling features of $\mathbf{F}_k^q$ and generates normalized features for cross-view data fusion.

Second, we normalize the original feature $\mathbf{F}_k^r \in \mathbb{R}^{C_r \times H_r \times W_r}$ of the reference image to generate $\hat{\mathbf{F}}_k^r \in \mathbb{R}^{C_r \times H_r \times W_r}$. It is necessary to preserve the spatial position information of $\mathbf{F}_k^r$ during the preprocessing step. Therefore, we apply L2 normalization to achieve this, as shown in the following:

$$\hat{\mathbf{f}}_k^r(m,i,j) = \frac{f_k^r(m,i,j)}{\sqrt{\sum_{l=1}^{C_q}\left(f_k^r(l,i,j)\right)^2}} \tag{5}$$

where $\mathbf{f}_k^r(m,i,j)$ and $\hat{\mathbf{f}}_k^r(m,i,j)$ represent the (m,i,j)th element of $\mathbf{F}_k^r$ and $\hat{\mathbf{F}}_k^r$, respectively. With (4) and (5), we generate the normalized features of the query and reference images.

Third, cross-view spatial attention is conducted to generate the weight of objects in the reference image based on the query image. We utilize the normalized features $\hat{\mathbf{F}}_k^q$ and $\hat{\mathbf{F}}_k^r$ as the query and key, respectively, and calculate an attention weight using the following definition:

$$\mathbf{S}_k = \mathrm{Norm}\left(\mathrm{Reshape}\left(\hat{\mathbf{F}}_k^q \cdot \hat{\mathbf{F}}_k^r\right)\right) \in \mathbb{R}^{H_r \times W_r} \tag{6}$$

where $\cdot$ represents an inner product operation and Reshape is used to reshape the weight into the desired size. Meanwhile, $\hat{\mathbf{F}}_k^q \cdot \hat{\mathbf{F}}_k^r$ denotes the similarity between the object in the query image and those in the reference image. To more effectively distinguish the target object from irrelevant ones, we also utilize a special normalization function to handle the attention weight. This normalization function enhances the weight of the target object while reducing the weight of irrelevant objects. In addition, it scales all weight values to the same range, mitigating the impact of value range differences. It is defined as follows:

$$s_k(i,j) = \frac{\hat{\mathbf{F}}_k^q \cdot \hat{f}_k^r(*,i,j) - \mathbf{s}_{\min}}{\mathbf{s}_{\max} - \mathbf{s}_{\min}}, \quad i \in [1, H_r], \ \ j \in [1, W_r] \tag{7}$$

where

$$\begin{aligned} \mathbf{s}_{\min} &= \min_{i,j} \hat{\mathbf{F}}_k^q \cdot \hat{f}_k^r(*,i,j) \\ \mathbf{s}_{\max} &= \max_{i,j} \hat{\mathbf{F}}_k^q \cdot \hat{f}_k^r(*,i,j). \end{aligned} \tag{8}$$

With (6), we obtain the attention weight between the query image and the reference image.

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Fourth, the fused feature $\mathbf{X}'_k$ is generated using the attention weight, which is defined as follows:

$$\mathbf{X}'_k = \mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv}\big(\hat{\mathbf{F}}^r_k \odot \mathbf{S}_k\big)\big)\big) \in \mathbb{R}^{C_r \times H_r \times W_r} \tag{9}$$

where Conv, BN, and ReLU represent convolutional networks, a batch normalization layer, and a nonlinear active function, respectively. $\mathbf{X}'_k$ is the fused feature generated by the query-aware cross-view fusion module using cross-view spatial attention. The fused feature $\mathbf{X}'_k$ pays more attention to the most relevant areas of the reference images that are most likely to contain the target object. With $\mathbf{X}'_k$, we can more effectively find the location of an object from cross-view images.

Finally, we apply a convolutional network on the fused feature $\mathbf{X}'_k$ to generate an embedding at each location for bbox prediction as follows:

$$\mathbf{X}_k = \mathrm{Conv}\big(\mathbf{X}'_k\big) \in \mathbb{R}^{D \times H_a \times W_a} \tag{10}$$

where $D = m \times (4+1)$. m is the count of anchor boxes, which are predefined bboxs to help identify objects with different sizes [64]; 4 denotes the dimension of the center coordinate, width, and height of a bbox; and 1 means that we use a 1-D embedding to predict whether a bbox is the target object or not.

With the fused features, bbox regression and classification are conducted in each position for generating a target bbox, i.e., the target geographic location. As shown in Fig. 5, bbox regression outputs the size and center coordinate of a bbox, and bbox classification predicts the probability that a query object is at a position.

E. Training Objective

`DetGeo` outputs a bbox and a confidence score indicating whether there is a query object at this position. The bbox is represented by a center coordinate, width, and height. We design a loss function to optimize `DetGeo`, which is defined as follows:

$$\mathcal{L} = L_{\mathrm{cls}} + \beta L_{\mathrm{geo}} \tag{11}$$

where L_{geo} and L_{cls} are a localization loss function and a classification loss function, respectively. The localization loss function is utilized to minimize the distance between the predicted bbox and the ground truth, whereas the classification loss function is employed to estimate the probability of an object presented in a particular grid cell or position. β is a hyperparameter and is typically set to 1. We employ the regression loss to implement L_{geo} as follows:

$$L_{\mathrm{geo}} = \sum_{k=1}^{n} \big(x_k - x_k^*\big)^2 + \big(y_k - y_k^*\big)^2 + \big(w_k - w_k^*\big)^2 + \big(h_k - h_k^*\big)^2 \tag{12}$$

where (x_k, y_k, w_k, h_k) denotes the predicted bbox of the kth instance. $(x_k^*, y_k^*, w_k^*, h_k^*)$ is the corresponding ground-truth bbox. In practice, we employ an anchor-based technique to minimize L_{geo}. Anchor boxes are a set of predefined bboxs of a certain height and width. In this approach, both the predicted and ground-truth bboxs are transformed into offsets with respect to a predefined anchor box. The localization loss is calculated over these offsets, which derives L_{geo} as follows:

$$\begin{aligned} L_{\mathrm{geo}} = \sum_{k=1}^{n} & \big(\sigma(x_k) - \big(x_k^* - \lfloor x_k^* \rfloor\big)\big)^2 \\ & + \big(\sigma(y_k) - \big(y_k^* - \lfloor y_k^* \rfloor\big)\big)^2 \\ & + \big(\log(w_k/w_a) - \log\big(w_k^*/w_a\big)\big)^2 \\ & + \big(\log(h_k/h_a) - \log\big(h_k^*/h_a\big)\big)^2 \end{aligned} \tag{13}$$

where $\sigma(\cdot)$ is a sigmoid function and $\lfloor \cdot \rfloor$ represents the operation of rounding down to the nearest integer. w_a and h_a denote the width and height of a selected anchor box, respectively. During the inference phase, the coordinates of the grid cell with the highest probability need to be added to both w_k and h_k to generate the actual object position. This operation transforms the offset predictions into absolute pixel coordinates.

L_{cls} is implemented by binary cross-entropy loss [65], which is defined as follows:

$$L_{\mathrm{cls}} = \sum_{k=1}^{n} o_k^* \log(o_k) + \big(1 - o_k^*\big) \log(1 - o_k) \tag{14}$$

where o_k is the predicted probability of the kth instance near the ground-truth position and o_k^* is the corresponding ground-truth label. When the anchor associated with the predicted bbox has the largest IoU with the ground-truth bbox, the corresponding o_k^* is assigned to 1, and all others are labeled as 0.

The optimization of `DetGeo` can be achieved based on $\mathcal{L}$. The best hyperparameter is obtained by grid search in this article. The detailed configuration of `DetGeo` and the experiment results will be discussed in Section V.

V. Experiment

A. Experimental Settings

1) Implementation Details: Our method is implemented in PyTorch. In this method, we use some operations to preprocess input data before data are fed into the network. For example, ground-view images, drone aerial-view images, and satellite-view images are resized to 256×512, 256×256, and 1024×1024, respectively. We apply data augmentation [66] to ground-view, drone aerial-view, and satellite-view images by perturbing them with translations, rotations, and so on. ResNet18 [61] and Darknet-53 [63] are used as the backbone for the query image encoder and the reference image encoder, respectively. ResNet18 and Darknet-53 have been pretrained on the ImageNet dataset. In addition, we generate anchor boxes via K-means clustering. The RMSProp optimizer is used to train the model with an initial learning rate of 0.0001. The learning rate decreases by half every 10 epochs. The batch size is 12 in all our experiments, and the total number of epochs is 25 for the CVOGL dataset.

2) Evaluation Settings: CVOGL aims to find a geographic location represented by a bbox. Thus, we apply Intersection over Union (IoU) to calculate the accuracy as our evaluation metric. The metric measures the overlap between two bboxs

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TABLE III
PERFORMANCE (acc%) COMPARISON WITH BASELINES AND OUR PROPOSED DetGeo ON THE CVOGL DATASET

Method	Ground → Satellite				Drone → Satellite			
	Test		Validation		Test		Validation	
	acc@0.25(%)	acc@0.5(%)	acc@0.25(%)	acc@0.5(%)	acc@0.25(%)	acc@0.5(%)	acc@0.25(%)	acc@0.5(%)
CVM-Net [16]	4.73	0.51	5.09	0.87	20.14	3.29	20.04	3.47
RK-Net [21]	7.40	0.82	8.67	0.98	19.22	2.67	19.94	3.03
L2LTR [19]	10.69	2.16	12.24	1.84	38.95	6.27	38.68	5.96
Polar-SAFA [27]	20.66	3.19	19.18	2.71	37.41	6.58	36.19	6.39
TransGeo [20]	21.17	2.88	21.67	3.25	35.05	6.37	34.78	5.42
SAFA [27]	22.20	3.08	20.59	3.25	37.41	6.58	36.19	6.39
DetGeo	**45.43**	**42.24**	**46.70**	**43.99**	**61.97**	**57.66**	**59.81**	**55.15**

by calculating the ratio of the intersection area to the union area of the two bboxs. It is defined as follows:

$$\text{acc@}t = \frac{1}{n}\sum_{i=1}^{n}\psi_i(t) \tag{15}$$

where

$$\psi_i(t) = \begin{cases} 1, & \text{if IoU}\left(b_i, b_i^*\right) \geq t \\ 0, & \text{else} \end{cases}$$

$$\text{IoU}\left(b_i, b_i^*\right) = \frac{\left|b_i \cap b_i^*\right|}{\left|b_i \cup b_i^*\right|}.$$

b_i and b_i^* are the predicted bbox and ground-truth bbox for the ith instance. $|b_i \cap b_i^*|$ and $|b_i \cup b_i^*|$ denote the overlap and union area between b_i and b_i^*. t is a threshold indicating whether the predicted bbox is correct or not. In this article, acc@0.25 and acc@0.5 are used to evaluate all methods.

B. Comparison With Baselines

We compare our DetGeo with the following cross-view image geo-localization methods on the CVOGL dataset.

1) **CVM-Net** [16] utilizes a Siamese neural network and a weighted soft margin ranking loss to acquire a viewpoint-invariant representation.
2) **SAFA** [27] applies a polar transform to warp satellite images, enabling alignment of ground- and satellite-view images geometrically.
3) **RK-Net** [21] jointly learns a discriminative representation and detects salient key points using a single network for cross-view geo-localization.
4) **L2LTR** [19] employs a Transformer to learn global dependencies of cross-view images, aiming to reduce visual ambiguities in cross-view matching.
5) **TransGeo** [20] proposes an attention-guided nonuniform cropping method to focus on crucial regions when learning cross-view representations using a Transformer.

In this article, we focus on CVOGL tasks, i.e., Ground → Satellite and Drone → Satellite. The research problem is a novel one, and there is no existing baseline. Existing cross-view image geo-localization methods mainly use image matching to find the location of an image. For all comparison methods, we divide the original satellite image into nonoverlapping patches with dimensions of 128 × 128 pixels. For example, if a satellite image has dimensions of 1024 × 1024 pixels, it is divided into 64 image patches, with each patch measuring 128 × 128 pixels. In addition, each image patch is distinct and does not overlap with any other patch. We retrieve the top-five similar patches as candidate locations for each query image. When calculating accuracy in (15), the boundaries of the retrieved patches are treated as the corresponding predicted bboxes. In addition, we calculate the IoU between the retrieved patch and the ground-truth bbox, and consider the largest IoU as the prediction result.

1) Performance of Cross-View Geo-Localization Models: The accuracy results for all baselines and our proposed method, DetGeo, are presented in Table III. We conducted a comprehensive comparison of DetGeo with state-of-the-art approaches on two CVOGL tasks: Ground → Satellite and Drone → Satellite. In the Ground → Satellite task, we use ground-view images as queries and satellite-view images as references to estimate the object locations. Similarly, in the Drone → Satellite task, we utilize drone aerial-view images as queries and satellite images as references to accurately identify the object locations.

Based on the results provided in Table III, we have made the following three observations. First, DetGeo demonstrates a significant performance advantage over other methods on the CVOGL dataset. Notably, in both the Ground → Satellite task and the Drone → Satellite task, DetGeo achieves an absolute improvement of 23.23% and 24.56%, respectively, compared to SAFA in terms of acc@0.25 on the test dataset. These results indicate that existing cross-view geo-localization methods may not be suitable for object geo-localization. Second, the accuracy of the baselines notably decreases as the IoU threshold increases. For instance, in the Ground → Satellite task, the acc@0.25 of SAFA in the test dataset is 22.20%, while its acc@0.5 drops significantly to 3.08%. These findings suggest that existing methods tend to provide coarse location estimation rather than precise object localization. Finally, DetGeo exhibits superior performance in the Drone → Satellite task compared to the Ground → Satellite task. For example, in the validation dataset, DetGeo achieves an acc@0.25 of 46.70% in the Ground → Satellite task, whereas it reaches 59.81% in the Drone → Satellite task. This discrepancy in performance can be attributed to the

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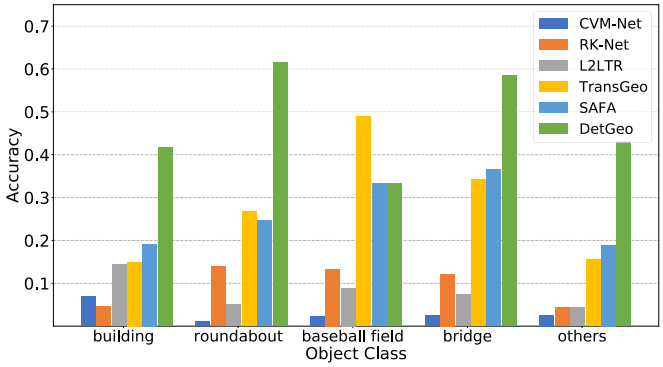


Fig. 7. Performance (accuracy%) across various object classes in the Ground → Satellite task.

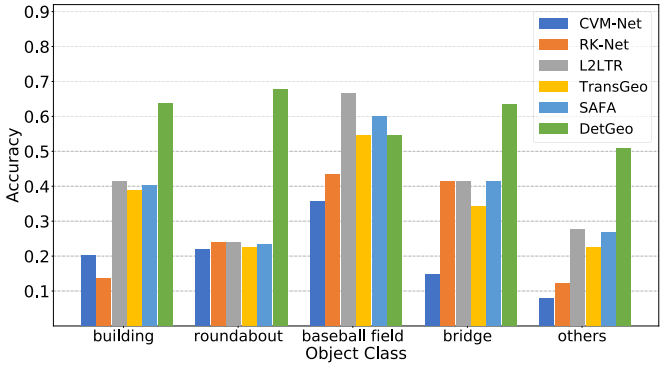


Fig. 8. Performance (accuracy%) across various object classes in the Drone → Satellite task.

larger domain gap between ground- and satellite-view images, in contrast to the drone- and satellite-view images. Overall, DetGeo outperforms other methods in both the Ground → Satellite and Drone → Satellite tasks on both the test and validation datasets.

2) Performance of Different Classes of Objects: We present the accuracy of all methods for some common object classes, as illustrated in Figs. 7 and 8 for the Ground → Satellite task and the Drone → Satellite task, respectively. From these figures, we can observe that DetGeo consistently achieves the highest performance across most classes in both tasks. Noteworthy, DetGeo outperforms other methods by a significant margin, especially in detecting roundabouts that are typically small objects in satellite images. This superiority is attributed to DetGeo's utilization of a detection-based framework, unlike other methods that rely on image patch matching-based approaches. DetGeo effectively leverages anchor-based techniques to find the locations of small objects, whereas other methods often generate large image patches and overlook small objects due to low IoU scores. Furthermore, due to the relatively smaller visual differences between drone- and satellite-view images compared to ground- and satellite-view images, most methods demonstrate superior performance in the Drone → Satellite task compared to the Ground → Satellite task. Overall, DetGeo consistently outperforms other methods across most object classes.

3) Performance of Different Sizes of Objects: Table IV presents the experimental results of our DetGeo method compared to other methods for objects of varying sizes using the acc@0.25 metric. The tasks "G → S" and "D → S" refer to the Ground → Satellite and Drone → Satellite tasks, respectively. Each column in the table corresponds to a specific size range, such as the values 1000 and 2000 indicating the size ranges (0, 1000] and (1000, 2000], respectively. From the results, we observe that our DetGeo method outperforms the other approaches significantly across various object sizes. For example, in the "G → S" task, DetGeo achieves an absolute improvement of 16.1% compared to SAFA for objects measuring 3000 m^2 in the test dataset. Similarly, in the "D → S" task, DetGeo shows an absolute improvement of 9.33% compared to SAFA for objects measuring 3000 m^2 in the test dataset. Furthermore, even when the object areas fall within the 1000 m^2 range, our method still performs impressively compared to the other approaches. This advantage can be attributed to our DetGeo framework, which effectively employs an anchor-based technique for the accurate localization of small objects. In contrast, the other methods rely on an image patch matching-based framework, which tends to filter out small objects more easily. Fig. 9 also illustrates examples of objects with different sizes. Overall, DetGeo consistently produces superior results compared to other methods across different object sizes.

TABLE IV
PERFORMANCE (ACCURACY%) ON DIFFERENT SIZES (m^2) OF OBJECTS

Task	Method	1,000	2,000	3,000	4,000	>4,000
G → S	CVM-Net [16]	1.75	8.45	4.24	6.76	3.02
	RK-Net [21]	4.55	8.11	9.32	10.81	8.04
	L2LTR [19]	6.29	12.84	16.10	13.51	9.55
	TransGeo [20]	12.94	22.30	28.81	28.38	24.12
	SAFA [27]	15.73	26.69	29.66	22.97	20.10
	DetGeo	**37.76**	**50.00**	**45.76**	**54.05**	**46.23**
D → S	CVM-Net [16]	12.59	22.97	31.36	28.38	17.09
	RK-Net [21]	11.19	18.58	35.59	21.62	21.11
	L2LTR [19]	26.57	42.57	60.17	48.65	35.18
	TransGeo [20]	20.98	41.89	52.54	39.19	33.17
	SAFA [27]	23.78	42.57	58.47	47.30	33.17
	DetGeo	**50.35**	**64.86**	**67.80**	**60.81**	**71.36**

4) Performance of Different Sizes of Regions: We evaluate all methods on regions of varying sizes using the acc@0.25 metric. The experiments are conducted on the test dataset. The size of regions includes 300 × 300, 512 × 512, and 1024 × 1024. Fig. 10 illustrates the results for the Ground → Satellite task, while Fig. 11 displays the results for the Drone → Satellite task. The x-axis in both figures represents the different sizes of regions. For example, 1024 × 1024 indicates a satellite image covers an area of 1024 × 1024 m^2. We observe that DetGeo outperforms other methods across all region sizes. This is because our method adopts a DetGeo framework, which effectively handles objects in regions of different sizes. In addition, we can find that most methods obtain the worst performance on the 1024 × 1024 regions. This is likely because these regions are commonly represented by low-resolution satellite reference images, which exhibit significant visual differences from query images. Nevertheless, even within these challenging regions,

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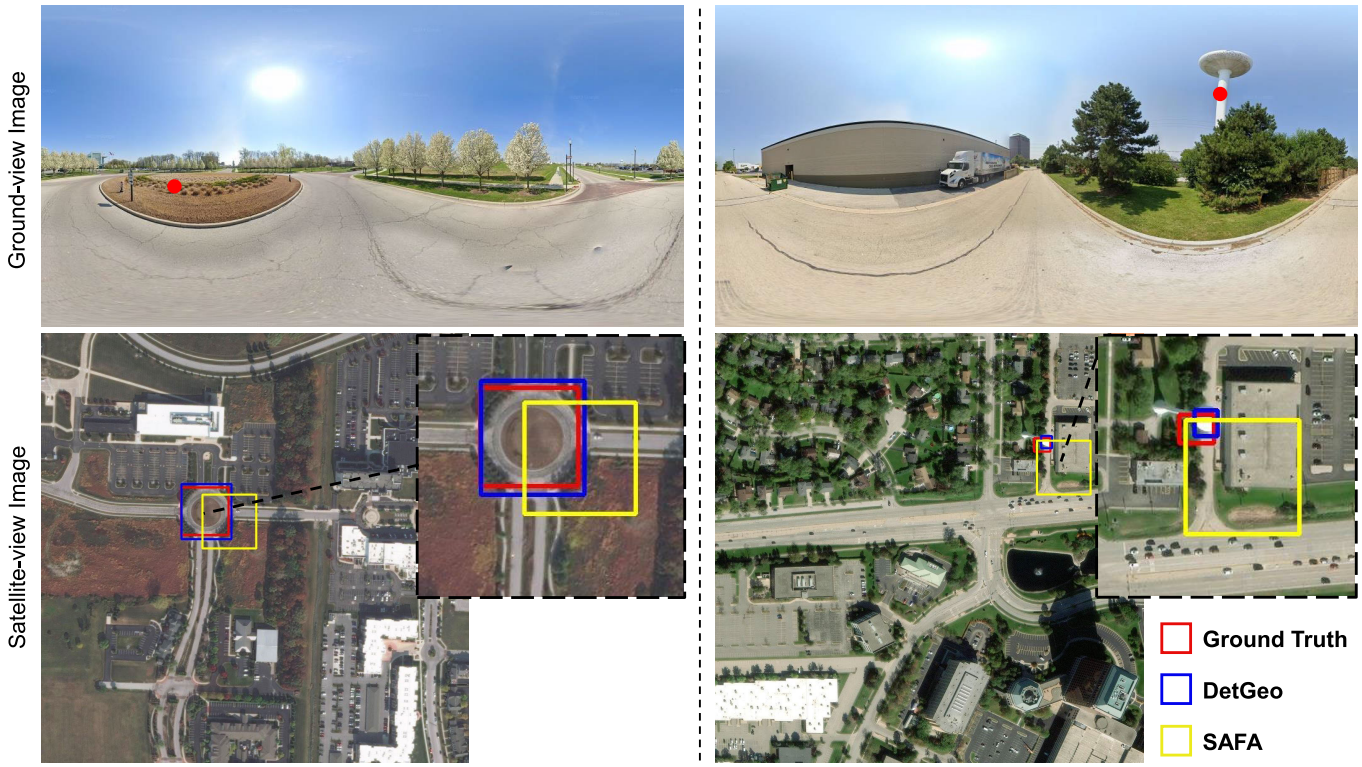


Fig. 9. Some results are produced by DetGeo and SAFA in the Ground → Satellite task.

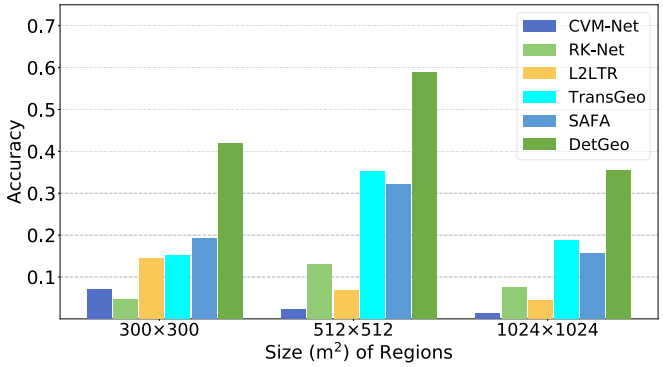


Fig. 10. Performance (accuracy%) across different sizes of regions in the Ground → Satellite task.

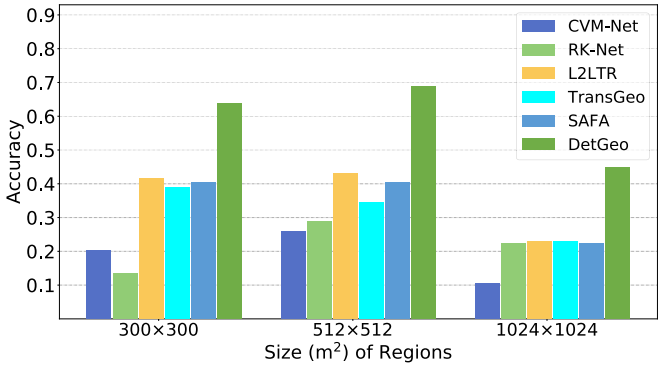


Fig. 11. Performance (accuracy%) across different sizes of regions in the Drone → Satellite task.

our method consistently achieves better results compared to other methods. Overall, DetGeo achieves higher accuracy across regions of different sizes.

C. Qualitative Analysis

We conducted a qualitative analysis on DetGeo and its baseline method SAFA, comparing their predicted results in Figs. 9 and 12 for the Ground → Satellite task and the Drone → Satellite task, respectively. In these figures, the ground-truth bboxs are shown in red, while the blue bboxs represent the predictions made by DetGeo and the yellow bboxs represent the selections made by SAFA. First, we observed that DetGeo can identify the precise location of query objects, whereas SAFA only provides a coarse approximation. For example, on the left-hand side of Figs. 9 and 12, the yellow bboxs produced by SAFA only encompass a portion of each roundabout. In contrast, DetGeo accurately detects the locations of both objects. Second, we can see that DetGeo can correctly localize objects of different sizes compared to SAFA. For example, on the right-hand side of Fig. 9, DetGeo successfully finds the location of a water tower, but SAFA filters it out due to a low IoU value. This is because DetGeo adopts a detection-based framework, while SAFA employs an image retrieval-based framework. Since objects in satellite images can vary significantly in size, the detection-based framework effectively addresses this challenge by utilizing an anchor-based technique. However, image retrieval-based methods struggle to find an optimal image division solution for accurately covering each object. In summary, DetGeo consistently outperforms existing image retrieval-based methods.

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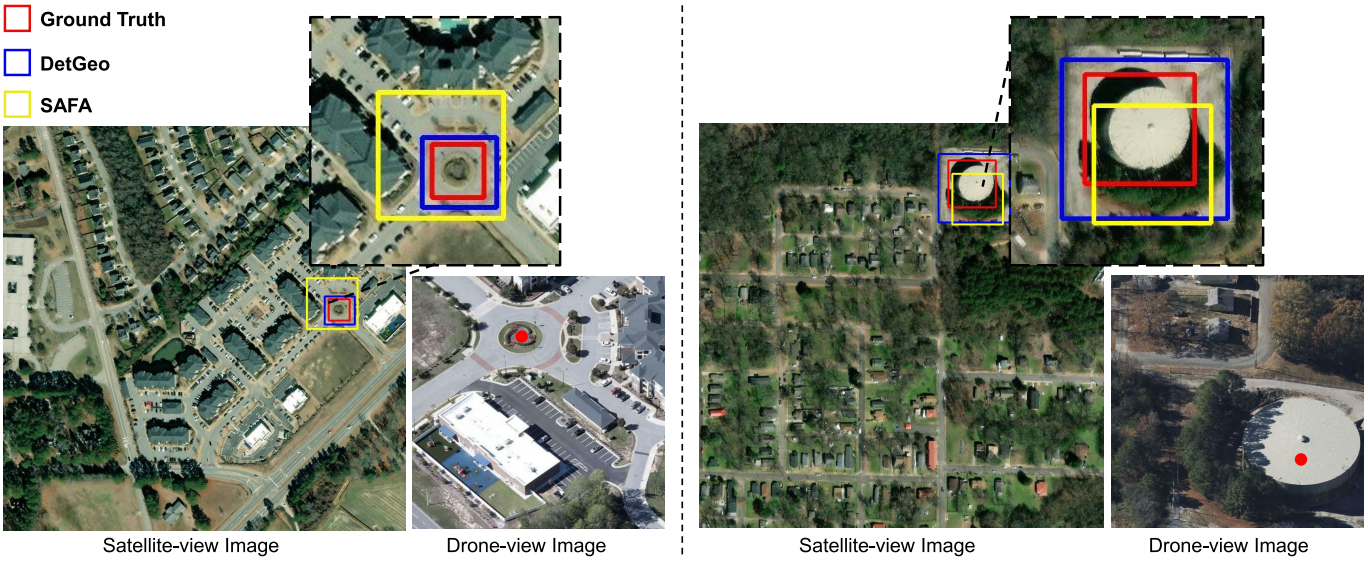


Fig. 12. Some results are generated by DetGeo and SAFA in the Drone → Satellite task.

TABLE V
PERFORMANCE (ACCURACY%) OF BASELINES ON DIFFERENT PATCH SIZES

Task	Method	Test		Validation	
		128×128	256×256	128×128	256×256
G → S	CVM-Net [16]	4.73	10.38	5.09	11.38
	RK-Net [21]	7.40	14.59	8.67	14.95
	L2LTR [19]	10.69	10.28	12.24	14.41
	TransGeo [20]	21.17	7.50	21.67	8.78
	SAFA [27]	22.20	17.68	20.59	19.61
D → S	CVM-Net [16]	20.14	10.28	20.04	12.68
	RK-Net [21]	19.22	23.23	19.94	24.59
	L2LTR [19]	38.95	25.08	38.68	27.52
	TransGeo [20]	35.05	24.88	34.78	26.44
	SAFA [27]	37.41	24.87	36.19	26.65

TABLE VI
ABLATION TEST RESULTS IN TERMS OF ACCURACY BY DISCARDING QUERY-AWARE CROSS-VIEW FUSION MODULE (W/O QCVFUSION)

Task	Method	Test		Validation	
		acc@0.25	acc@0.5	acc@0.25	acc@0.5
G → S	w/o QCVFusion	44.71	40.90	44.31	40.52
	DetGeo	45.43	42.24	46.70	43.99
D → S	w/o QCVFusion	44.91	41.42	48.10	44.31
	DetGeo	61.97	57.66	59.81	55.15

Furthermore, we analyzed the performance of different baselines on various patch sizes using the acc@0.25 metric, as shown in Table V. We observe that the accuracy of certain methods increases when the patch size is changed from 128 × 128 to 256 × 256. On the other hand, some methods experience a decrease in accuracy when the patch size is increased. For example, RK-Net demonstrates an absolute improvement of 7.19% and 4.01% in the "G → S" task and the "D → S" task, respectively. However, the accuracy of SAFA decreases from 22.20% to 17.68% in the "G → S" task and from 37.41% to 24.87% in the "D → S" task. These results indicate that the baselines are highly sensitive to the patch size. However, due to the size variation of object instances, it is not possible to find an optimal patch size to perfectly encompass each object in the satellite images. Thus, the baselines are less suitable than DetGeo for accurate object geo-localization.

Finally, we analyzed the query-aware cross-view fusion module, as depicted in Figs. 13 and 14. Satellite reference images often contain multiple objects and exhibit significant visual differences compared to query images. To focus on relevant objects, the fusion module utilizes cross-view spatial attention. We present the attention map using a pseudo-color image in Figs. 13 and 14. For example, in Fig. 13, a roundabout is selected in a ground-view image, while the corresponding area in the satellite-view image is highlighted. Similarly, in Fig. 14, the fusion module identifies and emphasizes a retrieved baseball field in the satellite image. These attention maps demonstrate that the fusion module pays more attention to important objects during the integration of cross-view images, resulting in more precise object geo-localization.

D. Ablation Study

To investigate the impact of different design choices in our proposed approach, we conducted an ablation study on the test and validation datasets, where we removed the query-aware cross-view fusion module (referred to as "w/o QCVFusion"). In the w/o QCVFusion method, we replaced the fusion module with a simple feature addition method to combine cross-view features. We reported the acc@0.25 (%) results and acc@0.5 (%) results of the ablation test in Table VI. The results indicate that the query-aware cross-view fusion module significantly enhances the performance of DetGeo, particularly in the "D → S" task. For example, in the "D → S" task, DetGeo outperforms the w/o QCVFusion method with absolute improvements of 16.24% and

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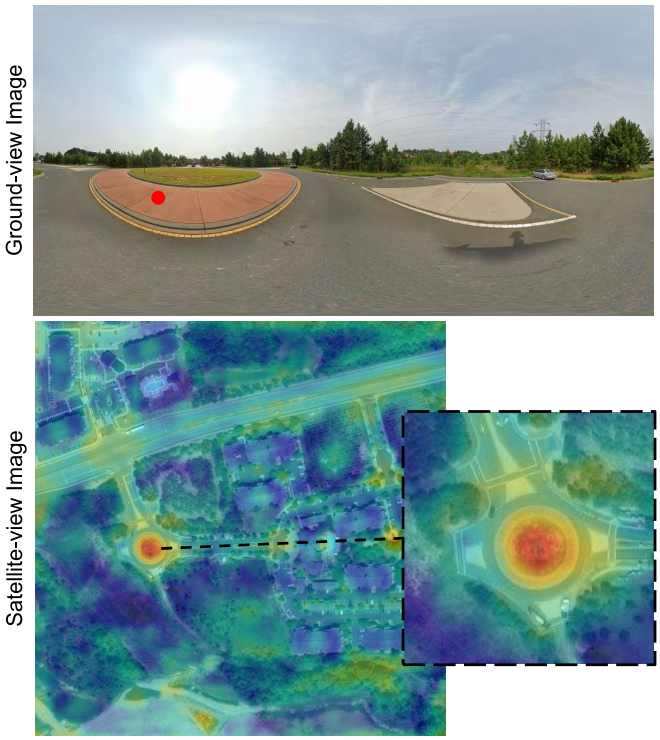


Fig. 13. Visualization of an attention map generated by the query-aware cross-view fusion module in the Ground → Satellite task. The first row displays a ground-view query image, while the second row shows a satellite-view reference image and an attention map.

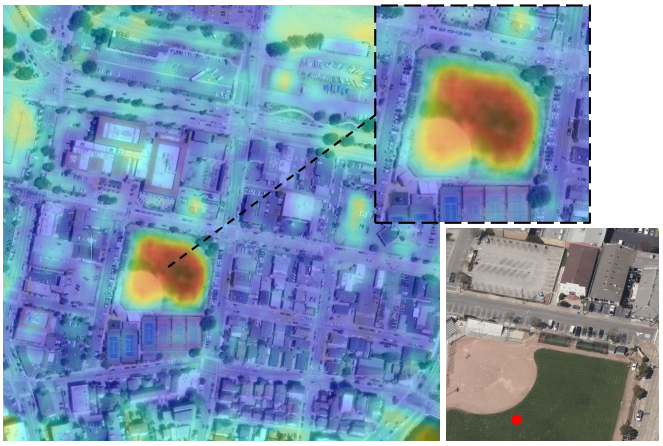


Fig. 14. Visualization of an attention map generated by the query-aware cross-view fusion module in the Drone → Satellite task. The left column shows a satellite-view reference image with an attention map, while the right column displays a drone aerial-view query image.

10.84% in terms of acc@0.5 on the test and validation datasets, respectively. Similarly, in the "G → S" task, DetGeo achieves increments of 3.28% and 8.56% in acc@0.5 on the test and validation datasets, respectively, compared to the w/o QCVFusion method. In summary, the query-aware cross-view fusion module in DetGeo significantly enhances the accuracy of object geo-localization.

E. Parameter Analysis

We conducted several experiments in the "G → S" task and the "D → S" task by varying β from 1 to 37. We reported the acc@0.25 metric for different β values on the validation dataset, as shown in Fig. 15. The hyperparameter β is used to balance the localization loss function and the classification loss function. Specifically, β controls the weight of the localization loss function, which aims to minimize the distance between the predicted bbox and the ground truth. We tested the accuracy at $\beta = 0.1$, which is considered an approximation of the accuracy at $\beta = 0$. We observed that the accuracy initially increases as β increases but gradually decreases beyond a certain threshold. This is because, when β is below a specific value, the localization loss and the classification loss are balanced. DetGeo can effectively estimate the probability of an object being at a certain location and its size. However, when β exceeds that value, the two losses become unbalanced, and DetGeo starts to prioritize location regression over the probability of an object. As a result, DetGeo struggles to accurately determine the position and size of an object. In addition, we can see that the performance of DetGeo in the "D → S" task is more sensitive to changes in β compared to the "G → S" task. This may be due to the larger visual differences between ground- and satellite-view images compared to drone- and satellite-view images. In the "G → S" task, DetGeo needs to pay more attention to the accurate location of identifying objects.

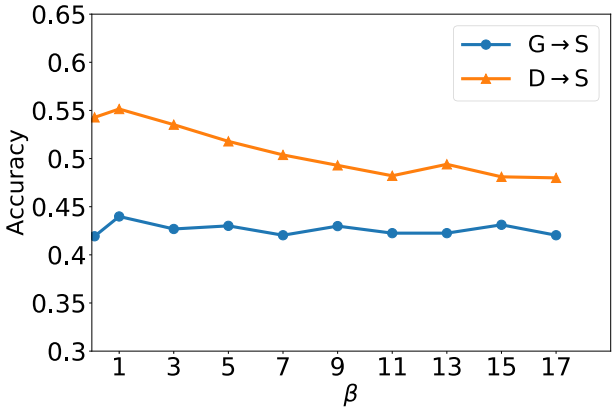


Fig. 15. Sensitivity analysis of β on CVOGL.

VI. Conclusion

In this article, we introduce a novel problem of CVOGL in a local region with high-resolution satellite images. This problem involves two similar geo-localization tasks: ground-to-satellite object localization and drone-to-satellite object localization. To tackle these tasks, we introduce a DetGeo method that utilizes an object detection-based framework with a two-branch encoder and a query-aware cross-view fusion module. The two-branch encoder comprises a query image encoder and a reference image encoder, extracting visual features from query images and reference images, respectively. The fusion module leverages cross-view spatial attention to effectively learn and combine information from cross-view images. In addition, we propose an objective function to train DetGeo effectively. To evaluate the performance of our method, we introduce a new CVOGL dataset. This dataset consists of ground- or drone-view images as query images

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and satellite images as geo-tagged reference images. Through extensive experiments conducted on the CVOGL dataset, we demonstrate that our method outperforms state-of-the-art methods significantly.

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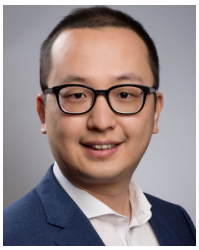
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